

KeratoScreen: Early Keratoconus Classification With Zernike Polynomial Using Deep Learning

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Purpose: We aimed to investigate the usefulness of Zernike coefficients (ZCs) for distinguishing subclinical keratoconus (KC) from normal corneas and to evaluate the goodness of detection of the entire corneal topography and tomography characteristics with ZCs as a screening feature input set of artificial neural networks.

Methods: This retrospective study was conducted at the Affiliated Eye Hospital of Wenzhou Medical University, China. A total of 208 patients (1040 corneal topography images) were evaluated. Data were collected between 2012 and 2018 using the Pentacam system and analyzed from February 2019 to December 2021. An artificial neural network (KeratoScreen) was trained using a data set of ZCs generated from corneal topography and tomography. Each image was previously assigned to 3 groups: normal (70 eyes; average age, 28.7 ± 2.6 years), subclinical KC (48 eyes; average age, 24.6 ± 5.7 years), and KC (90 eyes; average age, 25.9 ± 5.4 years). The data set was randomly split into 70% for training and 30% for testing. We evaluated the precision of screening symptoms and examined the discriminative capability of several combinations of the input set and nodes.

Results: The best results were achieved using ZCs generated from corneal thickness as an input parameter, determining the 3 categories of clinical classification for each subject. The sensitivity and

precision rates were 93.9% and 96.1% in subclinical KC cases and 97.6% and 95.1% in KC cases, respectively.

Conclusions: Deep learning algorithms based on ZCs could be used to screen for early KC and for other corneal ectasia during preoperative screening for corneal refractive surgery.

Key Words: keratoconus, Zernike polynomial, deep learning, corneal topography

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INTRODUCTION

The cornea provides approximately 70% of the eye's focusing power and is the most potent refractive element of the eye. Keratoconus (KC) is a progressive, noninflammatory thinning and conical protrusion of the cornea. It is often detected in teenage years to young adulthood and causes increasingly myopic and astigmatic vision.¹ Overall, 1% to 6% of the cases of subclinical KC (sub-KC) and clinical KC are detected through conventional screening before refractive surgery.^{2,3} Subclinical KC,^{4–6} the preclinical stage of KC, is difficult to detect in routine clinical practice. Early detection of KC is vital in the preoperative screening for refractive surgery candidates because surgery on an undiagnosed keratoconic cornea may result in iatrogenic keratectasia.⁷

Scheimpflug photography-based topography of the anterior and posterior surfaces of the human cornea is one of the most sensitive methods to detect KC.^{8,9} Information generated from these topographical maps help guide medical decision-making.⁵ Advanced KC can also be conventionally diagnosed based on biomicroscopic and keratometric measurements. A higher inferior–superior dioptric asymmetry (I-S) value indicates higher risk for developing KC.¹⁰ The subtle topographic changes in curvature, elevation, and corneal thickness presented in the early preclinical stage of the disease are crucial for its diagnosis.^{11,12} Although technological advances have enabled easier and more accurate detection of corneal ectasia, the discrimination of subclinical ectasia remains a challenge.^{13,14} In the earliest preclinical stage, subclinical ectasia will not show classical keratometric and biomicroscopic features. However, subtle topographic features similar to clinical KC can be observed on videokeratoscopy.¹³

Wave-front aberrations of the human cornea, which are based on an expansion of the Zernike polynomial (ZP), are commonly used in ophthalmology.^{15,16} It is a reliable and

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well-established method for evaluating corneal shape and structure by a series basis functions of sine and cosine.¹⁷ The phase content of these basis functions has various terms such as defocus, astigmatism, coma, and spherical aberrations. With a unit circle typically defined by the pupil aperture, the coefficients of ZP expansions can be used to analyze the development of corneal disease.^{15,17,18} ZPs are highly sensitive and specific in aiding KC diagnosis.^{15,17,18}

Machine learning has enormous potential in processing images and videos and has already been used for task classification based on image features. Machine learning algorithms have rapidly gained importance because of their high sensitivity and specificity in disease detection.^{19–21} Machine learning techniques are currently used in the screening of diabetic retinopathy,^{22,23} retinopathy of prematurity,²⁴ neovascular macular degeneration,²⁵ and uveitis.²⁶ Current research is directed toward the development of machine learning techniques for the early diagnosis of KC based on medical images.^{21,27–29}

Several machine learning methods, such as unsupervised learning,³⁰ convolutional neural networks (CNNs),^{20,31} ResNet-18,³² or support vector machine,³³ have been previously used for disease pathophysiology. CNN is one of the main methods used for recognizing and classifying ophthalmic images. However, most pathological images are of high resolution and are difficult for CNN to process directly. The publicly available ophthalmic pathology data sets are often limited, and the main problem with training a CNN model using a small data set is overfitting. The support vector machine views data points as *n*-dimensional vectors and separates such points with a (*n*–1)-dimensional hyperplane. Thus, it can be used for pattern classification. In this context, corneal topography has been used for automated pattern interpretation.^{30,34,35}

This study aimed to construct an artificial neural network (ANN)—KeratoScreen—using preconditioned Zernike coefficient (ZC) sets as the input for distinguishing sub-KC from normal eyes and KC eyes. This study also evaluated the performance of the multi-input modality ANN classifiers trained using topography data expressed with various ZCs. Corneal thickness was included as a parameter for evaluating the accuracy of classification with the ZC sets obtained from various maps of the corneas.

METHODS

Study Design and Population

This was a retrospective study of patients at the Eye Hospital of Wenzhou Medical University in Wenzhou, China. All patients were ethnically Chinese. This study was approved by the university's institutional review board and adhered to the tenets of the Declaration of Helsinki. The need for informed consent was waived because of the retrospective study design.

The corneal data of pachymetry and elevation were exported from the Pentacam HR system (Oculus, Optikgeräte GmbH, Wetzlar, Germany); a total of 208 eyes were enrolled to build the data set, and the eyes were divided into 3 groups. The inclusion criteria for eyes selected for the KC group were at least

one of the following on slitlamp biomicroscopy: Vogt striae, stromal thinning, Fleischer ring >2 mm arc, or corneal scarring, consistent with KC. The central average keratometry was more than 47.0 diopter (D). Asymmetric topographical features included I-S values¹⁰ ≥ 2.0 D of the vertical power gradient across the central 6-mm region. The subjects had no history of wearing contact lenses, ocular surgery, or extensive scarring.

The normal eye group included 70 normal subjects (70 eyes; average age, 28.7 ± 2.6 years) and was selected using the following criteria: no clinical symptoms of KC or subclinical KC in either eye, the central average keratometry was <45.0 D, the I-S value of corneal topographical features was <1.4 D at the vertical power gradient across the 6-mm region, the spherical equivalent was within ± 6.0 D, and astigmatism was within 2 D. The subjects also had no ocular disease, no prior ophthalmic surgery, and no cornea ectasia on a slitlamp. Only 1 eye was randomly selected from normal subjects for analysis.

The subclinical KC group, including patients with sub-KC (48 eyes; average age, 24.6 ± 5.7 years), was selected from the patients with undiagnosed eyes of unilateral KC, based on the following criteria: 1 eye was diagnosed with KC; in the contra, there were no clinical symptoms of KC during a slitlamp examination, retinoscopy, or ophthalmoscopy; the central average keratometry in this eye group was <45.0 D; the I-S value of the corneal topographical features was <1.4 D at the vertical power gradient through the central 6-mm region; the spherical equivalent was within ± 6.0 D; astigmatism was within 2 D; and these subjects had no history of contact lens wear or ocular surgery. The KC group included patients with KC (90 eyes; average age, 25.9 ± 5.4 years). Clinical KC was defined according to a combination of diagnosed characteristics of KC with topographic signs—including both changes on posterior corneal surface and corneal thickness progression, central keratometry more than 47.0 D, and I-S values more than 2.0 D of the vertical power gradient across the 6-mm region.³⁶ The demographic characteristics of 3 groups are given in Table 1.

Data consisted of sagittal curvature maps (anterior surface and posterior surface), elevation maps (anterior surface and posterior surface), and corneal thickness for a

TABLE 1. Demographic Characteristics of 3 Groups

	Normal (n = 70)	Sub-KC (n = 48)	KC (n = 90)
Age (y)	28.7 ± 2.6	24.6 ± 5.7	25.9 ± 5.4
Sex (F/M)	31/39	16/32	29/61
SE (D)	-4.6 ± 2.1	$-3.5 \pm 2.5^*$	$-6.3 \pm 4.1^*$
CDVA (decimal VA)	1.1 ± 0.1	$1.0 \pm 0.2^\dagger$	$0.6 \pm 0.3^\dagger$
Min-K (D)	42.5 ± 1.5	43.1 ± 1.9	$47.7 \pm 4.8^\dagger$
Max-K (D)	43.8 ± 1.6	44.6 ± 1.3	$50.2 \pm 5.6^\dagger$
Avg-K (D)	43.2 ± 1.5	43.9 ± 2.1	$47.6 \pm 4.8^\dagger$
Ast-K (D)	1.3 ± 0.7	$1.6 \pm 0.9^\dagger$	$3.8 \pm 2.4^\dagger$

**P* < 0.05 unpaired Student *t* test compared with the normal group.

[†]*P* < 0.05 Mann–Whitney *U* test compared with the normal group.

Normal, normal group; SE, spherical equivalent; CDVA, corrected distance visual acuity; Max-K, maximum keratometry; Ast-K, astigmatic keratometry; Avg-K, average keratometry; D, diopter; Min-K, minimum keratometry; VA, visual acuity.

range of ± 7 mm around the corneal apex in 0.1-mm increments with minimum corneal thickness. The data were exported to the Pentacam system as CSV files in the form of 141×141 anterior and posterior matrices. Figure 1 shows the topographic pattern from representative cases of 3 categories in this study. Figures 1A–C show the anterior corneal curvature matrices visualized in heat maps with contour for normal eye, sub-KC eye, and KC eye, respectively.

All topography maps generated by the Pentacam system were originally based on elevation data derived from Scheimpflug images and the 3D model of the anterior eye segment reconstructed from those images. The sagittal and tangential radii were calculated directly from the elevation data.

Corneal Surface Representation Using Zernike Polynomial

Different expansion coefficients for the same wave-front error are inferred by different pupil sizes. Thus, to

generate comparable input ZC sets between groups, we first normalized all the maps through min–max normalization using Equation (1).

In general, the ZPs are defined as follows:

$$Z_n^m(\rho, \theta) = \begin{cases} N_n^m R_n^{|m|}(\rho) \cos m\theta & \text{for } m \geq 0, \\ -N_n^m R_n^{|m|}(\rho) \sin m\theta & \text{for } m < 0, \end{cases} \quad (1)$$

where the normalization factor is N_n^m and the radial polynomial is $R_n^{|m|}$,

$$N_n^m = \sqrt{\frac{2(n+1)}{1+\delta_{m0}}}, \quad (2)$$

$$R_n^{|m|}(\rho) = \sum_{s=0}^{(n-|m|)/2} \frac{(-1)^s (n-s)}{s! [0.5(n+|m|)-s]! [0.5(n-|m|)-s]!} \times \rho^{n-2s}, \quad (3)$$

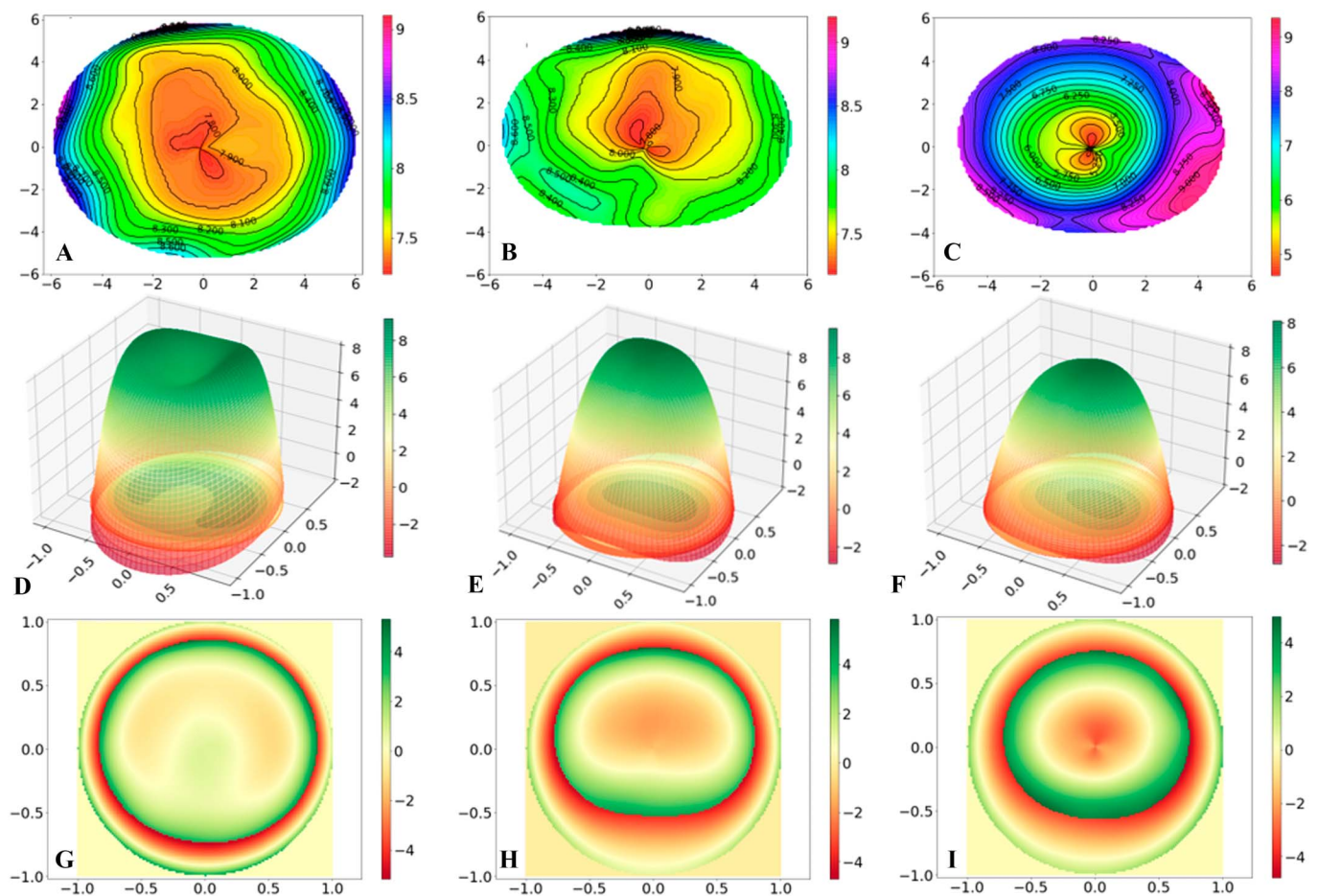


FIGURE 1. Example maps generated from the anterior corneal curvature. A, D, and G are from the same eye of the normal group. B, E, and H are from the same eye of the sub-KC group. C, F, and I are from the same eye of the KC group. A, B, and C are heat maps with contour. D, E, and F represent anterior corneal surface fitting as a function of ZCs. G, H, and I are remaining aberrations in 2D. (The full color version of this figure is available at www.corneajrnl.com.)

where δ_{m0} is the Kronecker delta function.

Thereafter, we used an algorithm to rescale the pupil size to a normalized radius based on ZP fitting as shown in Equation (4), previously described by Schwiegerling^{16,37,38} and Campbell.³⁸

The polar coordinates (r, θ) must therefore be scaled to the normalized polar coordinates (ρ, θ) by setting $\rho = \frac{r}{r_{pupil}}$; the surface $S(r, \theta)$ can be represented as a linear combination of Zernike terms,

$$S(r, \theta) = \sum_n \sum_m a_{n,m} z_n^m \left(\frac{r}{r_{pupil}}, \theta \right), \quad (4)$$

where $a_{n,m}$ are the expansion coefficients extent of the pupil radius and r_{pupil} is the value of the different pupil size.

Figures 1D to 1I show the images generated from the anterior corneal surfaces due to the Zernike fitting, Figures 1D to 1F display the 3D surface maps, and Figures 1G to 1I show the 2D surfaces after removal of fitting aberrations. Figures 1A, D, and G are derived from the same normal eye map in the data set; Figures 1B, E, and H are obtained from the same sub-KC eye; and Figures 1C, F, I are obtained from the same KC eye.

Network Architecture

Figure 2 shows the KeratoScreen schematic with a preprocessing model network architecture. The processing model (Fig. 2A) in KeratoScreen is described in brief as follows: The preprocessing model separately established ZC data sets of 5 corneal surface maps—the anterior corneal curvature (DA), posterior corneal curvature (DP), anterior corneal elevation (DAE), posterior corneal elevation (DPE), and corneal thickness data (pachymetry DPAC). The first N ($N \leq 15$) ZC terms (5 orders, when $N = 15$) obtained from each map were used to form ZC data sets.

KeratoScreen (Fig. 2B), with $L = 4$ layers (1 input layer, 2 hidden layers, and 1 output layer), was used in this study. The neurons on the input layer have N nodes that take an input vector $ZC = (ZC_0, ZC_1, \dots, ZC_{N-1})$ and outputs through 2 hidden layers. The composite map is defined as follows:

$$z^{(l+1)} = f^{(l+1)}(w_l^T z^{(l)} + b_l) \quad (l = 0, \dots, L-1), \quad (5)$$

where $f^{(l+1)}$ is given univariate activation link functions; $f^{(1)}$, $f^{(2)}$ are Rectified Linear Unit (ReLU) functions; $f^{(3)}$ is the soft-max function; w_l is an $N^{(l)} \times N^{(l+1)}$ parameter matrix; the superscript T means matrix transpose; b_l is an $N^{(l+1)}$ -dimensional parameter vector; $z^{(0)} = ZC$; $y = z^{(L-1)}$; and $N^{(l)}$ is the number of neurons on the l th layer.

Training

The data sets were used to train the ANN, with the number of neurons in the input layer set as a variable ranging from 1 to 15 corresponding to the number of ZCs. The last layer classified the category using a soft-max classifier with 3 neurons. The deep neural network was implemented in the

Python programming language using Keras 2.2.4 and TensorFlow 2.4.0 libraries.

The ZC data sets were randomly split into 2 groups: 70% for training and 30% for testing. A dropout method was used in the training process to prevent overfitting, and 30% of the neurons, which were selected at random, were ignored during the training phase. The model will overfit the training data set and have poor performance in the test set with too much training³⁹; thus, early stopping was used to avoid overtraining. In our case, the training loss after each epoch was monitored; if the test accuracy did not improve after 50 epochs, training was interrupted. The output layer created 3 output values because this was a 3-class classification problem, one for each class. The output value with the most considerable rate was considered as the class predicted by the model.

The parameters in the above model $w_1, w_2, w_3, b_1, b_2,$ and b_3 are estimated by minimizing the categorical cross entropy, a cross-entropy loss function for our multiclass classification problem specifies as follows: let $z^{(3)} = p$, $p = [p_1, p_2, p_3]$, and $\sum_{i=1}^3 p_i = 1$. We used multiple categorical variables to correspond to the 3 groups, each group with values of 1, 2, and 3, defined as follows:

$$d_i = \begin{cases} 1 & \text{if Normal group} \\ 2 & \text{if Sub-KC group} \\ 3 & \text{if KC group} \end{cases} \quad (6)$$

If the categorical group variable corresponded to d_i , we defined the one-hot encoding to be e_{ij} , where $e_{ij} = 1$ if $i = j$ or else it was 0. One-hot encoding replaces a single categorical group variable with 3 binary variables. The loss function is defined as follows:

$$Loss = \sum_{k=1}^3 e_{jk} \log\left(\frac{1}{p_k}\right) = \log\left(\frac{1}{p_j}\right). \quad (7)$$

To assess the performance of the algorithm, we collected the results of multiple experiments and used the mean to summarize the skill of our model. In our study, we used 20 repeats with the randomly generated training and test sets, and the results were averaged.

Statistical Analysis

Statistical analysis used SPSS 24.0 (SPSS for Windows, SPSS Inc). Data for continuous features are presented as mean $\pm$ SD. The Shapiro–Wilk test was used to assess the normality of data distribution. The Mann–Whitney U test, unpaired Student t test, or χ^2 test was used to compare corneal metrics among subclinical KC, KC, and normal subjects. P values < 0.05 were considered statistically significant.

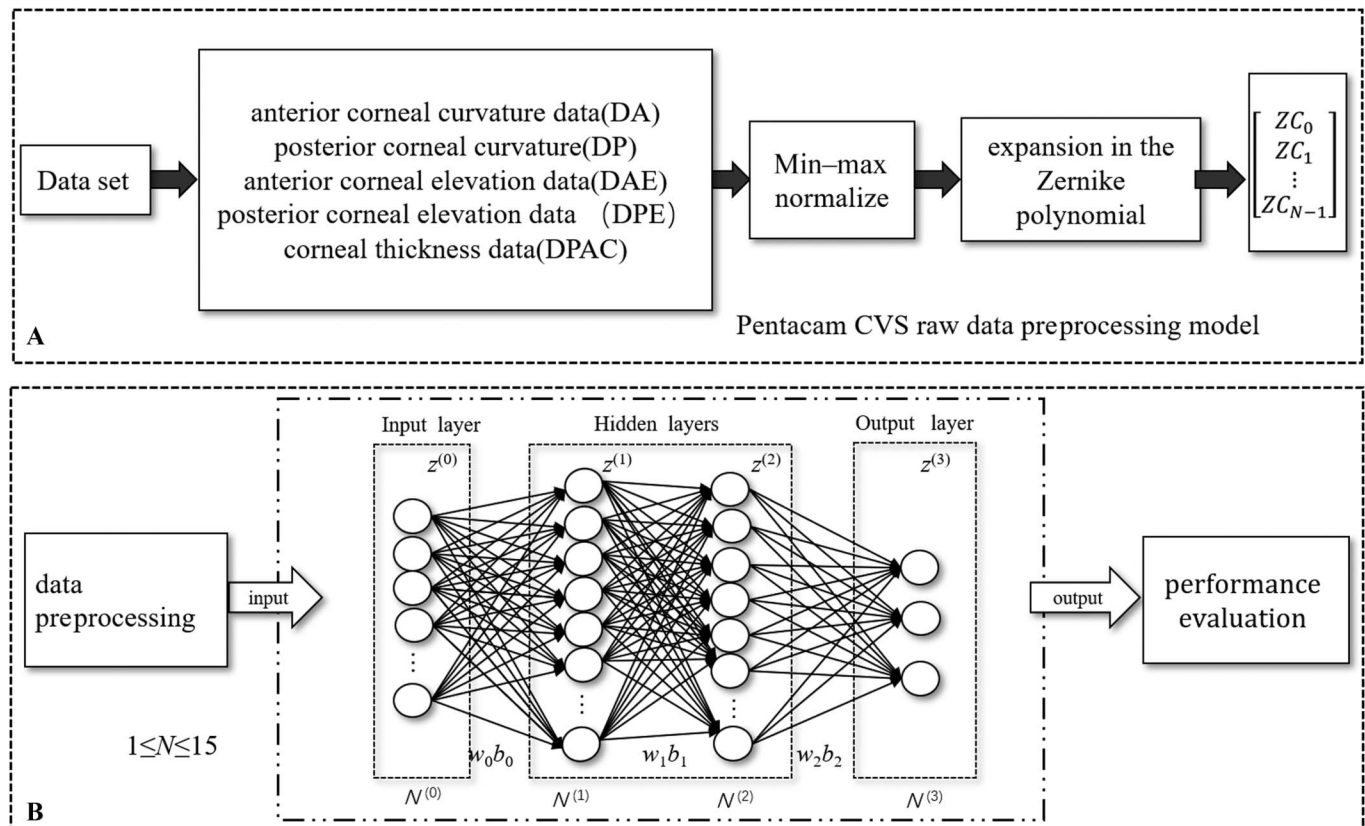


FIGURE 2. Schematic of the KeratoScreen architecture with preprocessing. A: Performance evaluation model. B: Network architecture.

RESULTS

The input sets were composed of individual ZCs generated from the corneal thickness and had the highest discriminative accuracy (DPAC). Figure 3 shows the average accuracy in the classification of eyes in the 3 groups with a different number of ZCs obtained in the anterior surface,

posterior surface, anterior elevation, posterior elevation, and corneal thickness. Notably, such high accuracy was obtained when the input set consisted of 6, 8, and 15 ZC terms from anterior corneal curvature (DA). Furthermore, when the input set consisted of 15 ZC terms, all ZC data—DA, DPE, DP, DAE, and DPAC—achieved a higher accuracy rate.

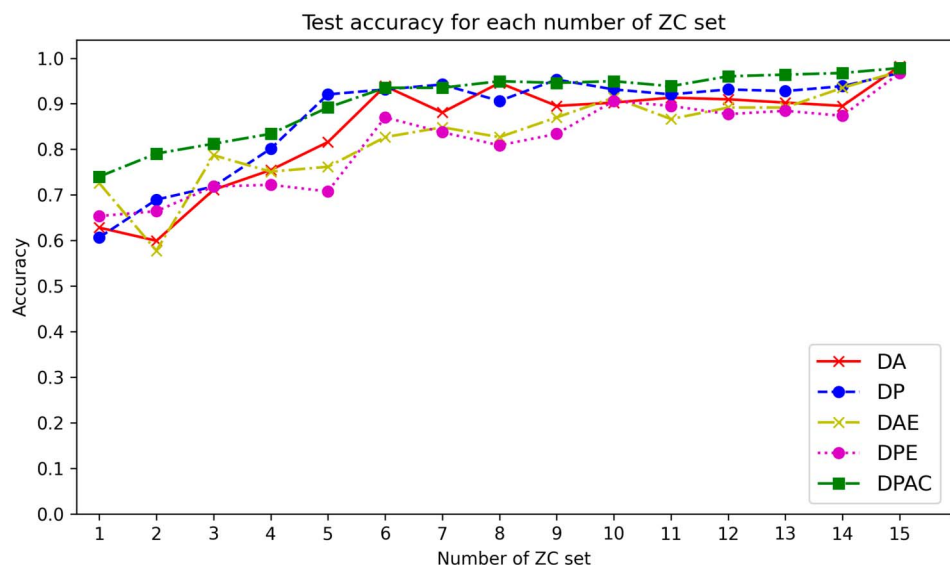


FIGURE 3. Classification accuracy as a function of ZC number for the 5 categories. (The full color version of this figure is available at www.corneajrnl.com.)

TABLE 2. Results of Screening Accuracy of the ZC for Different Orders

Order Coefficients	Group	DA	DP	DAE	DPE	DPAC
1 order	Normal	64.8%	58.2%	89.0%	74.7%	81.3%
	Sub-KC	52.1%	53.1%	60.4%	60.4%	57.3%
	KC	72.2%	71.1%	68.9%	61.1%	84.4%
2 orders	Normal	70.3%	74.7%	81.3%	83.5%	83.5%
	Sub-KC	62.5%	66.7%	69.8%	67.7%	72.9%
	KC	81.1%	74.7%	85.6%	64.4%	87.8%
3 orders	Normal	90.1%	94.5%	83.5%	94.5%	92.3%
	Sub-KC	95.8%	91.7%	76.0%	77.1%	91.7%
	KC	94.5%	93.3%	88.9%	90.0%	96.7%
4 orders	Normal	87.9%	93.4%	90.1%	91.2%	94.5%
	Sub-KC	90.6%	91.7%	89.6%	88.5%	92.7%
	KC	93.4%	94.4%	94.4%	92.2%	97.8%
5 orders	Normal	97.8%	98.9%	96.7%	94.5%	95.6%
	Sub-KC	97.9%	93.8%	94.8%	95.8%	97.9%
	KC	98.9%	97.8%	100.0%	100.0%	100.0%

DA, anterior corneal curvature; DP, posterior corneal curvature; DAE, anterior corneal elevation; DPE, posterior corneal elevation; DPAC, corneal thickness.

Table 2 summarizes the accuracy of the ZC for different orders, which shows the goodness of prediction of discrimination by measuring the ZCs for various corneal maps. The accuracy of all categories ranged from 93.8% to 100% by using 5 orders, while the highest accuracy was achieved 100% in the KC group (DAE, DPE, and DPAC) and the lowest was 93.8% in the sub-KC (DP). The input set comprising 1 order had the lowest accuracy at 52.1% (DA) in the sub-KC group. Overall, more ZC orders were added, the accuracy of 3 categories improved, and the result approached the maximum accuracy found with 5 orders.

Table 3 summarizes the classification performance of the test sets obtained using input ZCs of the 5 map categories: DA, DPE, DP, DAE, and DPAC. All categories revealed very comparable performances, with a validation accuracy ranging from 88.1% to 95.5%. The test accuracies of DPAC, DP, DA, DAE, and DPE were 95.5%, 93.3%, 91.7%, 89.3%, and 88.1%, respectively. DPAC showed slightly better performance than the other map categories in average precision of the KC group. Overall, the sensitivity and precision were inferior to those of the DPE of sub-KC.

DISCUSSION

There are concerns about reliability, precision, and accuracy when using deep learning models based on ZCs to screen for early-stage KC. Our study provides a better understanding of the relative usefulness of ZC counts for screening with different corneal zones, including corneal curvature, corneal elevation, and pachymetry. A significant accuracy in overall screening performance was observed using only a small number of ZCs. Our results showed a saturation state when the input set consisted of few ZCs, the addition of 6 to 9 high-order terms improved the overall accuracy. However, the addition of more than 9 terms had a lower impact on the classification accuracy. This result is consistent with that of previous reports in relation to the orders of the fit to cornea.^{17,40} Similarly, the number of terms matched the order of polynomials used as inputs of KeraToScreen, and the same classification accuracy trend was also observed. For example, the model developed with radial orders 0, 1, 2 (C_0^0, C_1^1, C_1^{-1}) achieved 71.6% accuracy in the KC group and that with radial orders 0, 1, 2 ($C_0^0, C_1^1, C_1^{-1}, C_2^{-2}, C_2^0, C_2^2$) achieved a significant improvement 92.7% accuracy in the KC group. However, the model with radial orders 0, 1, 2, 3 ($C_0^0, C_1^1, C_1^{-1}, C_2^{-2}, C_2^0, C_2^2, C_3^{-3}, C_3^1, C_3^3, C_3^{-3}$) had a slightly higher accuracy of 94.5%.

Several studies have assessed the usefulness of machine learning in distinguishing eyes with KC from normal eyes.^{34,35,41} Numerous algorithms, such as aberrometry parameters combined with ZCs and additional metrics based on epithelial thickness mapping, have also been studied to describe the properties of corneal topography and tomography in both KC and sub-KC eyes.^{42–44} Carvalho et al⁴³ reported the efficiency of their neural network systems, created using ZCs as an input, to diagnose KC eyes. However, the optimal number of ZCs in neural networks needed to accurately screen for KC, sub-KC, and normal eyes remains unclear.

In this study, we compared the classification performance of the ANN KeraToScreen by inputting the ZC sets derived from corneal curvature, elevation, and pachymetry individually. Corneal ectasia progression was defined based on steepening of the anterior corneal surface and steepening of the posterior corneal surface, with gradual increases in corneal thickness from the center to the periphery.³⁶ We found that ZCs representing corneal pachymetry achieved the

TABLE 3. Results of the Different Maps

	Normal			Sub-KC			KC			Total
	Sensitivity	Precision	F1 Score	Sensitivity	Precision	F1 Score	Sensitivity	Precision	F1 Score	Accuracy
DA	90.4%	90.1%	90.2%	91.7%	92.6%	92.1%	92.9%	92.4%	92.7%	91.7%
DP	95.5%	97.8%	96.6%	90.9%	90.5%	90.7%	93.8%	92.1%	92.9%	93.3%
DAE	92.4%	88.2%	90.3%	82.6%	87.5%	85.0%	93.3%	92.6%	93.0%	89.3%
DPE	92.2%	86.8%	89.4%	81.6%	86.3%	83.9%	91.0%	91.8%	91.4%	88.1%
DPAC	95.2%	95.5%	95.3%	93.9%	96.1%	95.0%	97.6%	95.1%	96.3%	95.5%

F1 score is the harmonic mean of precision and recall. Sensitivity is the fraction of the true labels that are correctly identified for each class. Precision is the fraction of the predicted labels that are correct for each class. The accuracy is the fraction of correctly identified pixels.

highest accuracy, possibly indicating that the corneal thickness varies during the early stage of KC.⁴⁵ Moreover, ZCs based on the corneal surface curvature had the second-highest predictive capability possibly because the morphologic distortion of the corneal shape is associated with KC.

Furthermore, the anterior corneal curvature functions better than the posterior curvature. The curvature maps are calculated from elevation data. The features of the anterior corneal curvature are the primary basis for detecting corneal abnormalities.^{42,46} ZP can be used for characterizing wavefront aberrations of the complex corneal shapes. ZCs of the corneal surface maps are highly sensitive and specific for detecting sub-KC. ZC data from curvature maps can reveal subtle changes of corneal ectasia, and they will be slightly better than the corneal elevation maps. ZCs derived from anterior corneal elevation demonstrated better performance than ZCs derived from posterior corneal elevation.

Our findings are consistent with those of previous studies on the applications of machine learning to discriminate between KC and healthy eyes. These studies used corneal biomechanical indices and tomographic indices as an input,^{19,30,31,35,47} and the accuracy ranged from 79.00% to 96.56%. KeratoScreen had an average accuracy of 97.4% by using 5 orders as an input. Most studies on KC used non-machine learning approaches.^{28,42,46,48} Notably, KeratoScreen showed better performance for classifying normal, sub-KC, and KC. The performance of each classification is described in detail below.

A normal cornea maintains a smooth and stable curvature.⁴⁹ For normal eye classification, all ZCs based on 5 corneal categories showed good performance. The classifier class recall was 95.5%; precision, 97.8%; and F1 score, 96.6%. Such a high F1 score indicates a good balance between precision and recall. This explains the minimal screening error in normal eyes for all corneal data. For sub-KC eyes, the precision analysis showed that KeratoScreen screening with corneal thickness data outperformed the screening with other type of data; 96.1% of true sub-KC eyes can be correctly identified with KeratoScreen. Several studies used the change in corneal thickness to discriminate between KC and healthy eyes.^{50,51} A higher magnitude of ZP indices was found to indicate more severe KC.⁴⁵ This result suggests that for accurate identification of sub-KC, the input ZC should be calculated based on corneal thickness data, and at least 6 ZCs should be used. For the classification of KC eyes, the KeratoScreen recall rate was 97.6% and the highest precision rate was 95.1%. Furthermore, the F1 score reached 96.3% when the ZC was based on corneal thickness. This indicates that KeratoScreen can correctly identify most of the true KC eyes; however, it should be noted that few non-KC eyes were also classified as KC. For clinical purposes, the identification of a patient with sub-KC is more important than the identification of KC eyes because corneas with undetected KC are known to be at high risk for iatrogenic keratectasia. Our result also suggests that corneal thickness data are suitable for screening patients with KC, provided that 6 to 15 ZCs are used as an input set.

This study had some limitations. First, we included a relatively small sample size of patients with sub-KC. An

ANN's capability depends on its training for accurate response, and the results will also decrease significantly when there are inadequate data in a given class. It will also benefit greatly from external validation with an external database as well. The second limitation is the poor interpretability of an ANN. More complicated deep learning methods need to be investigated to find more sensitive indices of sub-KC. Furthermore, given that deep learning program is limited by the quality of the data set used to train and validate it, more data will ensure better results.

CONCLUSIONS

Our neural network KeratoScreen may be a useful tool in diagnosing suspected KC. ZCs generated from corneal thickness as an input achieved the highest performance in the detection of sub-KC. ZCs based on corneal topography surface elevation data and thickness data are highly accurate in identifying KC, sub-KC, and normal KC. However, applying more than 9 terms did not significantly improve the accuracy of classification. KeratoScreen achieves the highest accuracy when the input set contains 15 ZCs extracted from the DPAC.

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REFERENCES

1. Krachmer JH, Feder RS, Belin MW. Keratoconus and related non-inflammatory corneal thinning disorders. *Surv Ophthalmol*. 1984;28:293–322.
2. Finis D, Ralla B, Karbe M, et al. Comparison of two different scheimpflug devices in the detection of keratoconus, regular astigmatism, and healthy corneas. *J Ophthalmol*. 2015;2015:315281–315286.
3. Binder PS, Lindstrom RL, Stulting RD, et al. Keratoconus and corneal ectasia after LASIK. *J Cataract Refract Surg*. 2005;31:2035–2038.
4. Martínez-Abad A, Piñero DP. New perspectives on the detection and progression of keratoconus. *J Cataract Refract Surg*. 2017;43:1213–1227.
5. de Sanctis U, Loiacono C, Richiardi L, et al. Sensitivity and specificity of posterior corneal elevation measured by pentacam in discriminating keratoconus/subclinical keratoconus. *Ophthalmology*. 2008;115:1534–1539.
6. Fukuda S, Yamanari M, Lim Y, et al. Keratoconus diagnosis using anterior segment polarization-sensitive optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2013;54:1384–1391.
7. Rabinowitz YS. Keratoconus. *Surv Ophthalmol*. 1998;42:297–319.
8. Henriquez MA, Hadid M, Izquierdo L. A systematic review of sub-clinical keratoconus and forme fruste keratoconus. *J Refract Surg*. 2020;36:270–279.
9. Kataria P, Padmanabhan P, Gopalakrishnan A, et al. Accuracy of scheimpflug-derived corneal biomechanical and tomographic indices for detecting subclinical and mild keratectasia in a South Asian population. *J Cataract Refract Surg*. 2019;45:328–336.
10. Li X, Rabinowitz YS, Rasheed K, et al. Longitudinal study of the normal eyes in unilateral keratoconus patients. *Ophthalmology*. 2004;111:440–446.

11. LJ M, WM B. Corneal topography of early keratoconus. *Am J Ophthalmol*. 1989;108:107–112.
12. Rabinowitz YS, Garbus J, McDonnell PJ. Computer-assisted corneal topography in family members of patients with keratoconus. *Arch Ophthalmol*. 1990;108:365–371.
13. Saad A, Gatinel D. Topographic and tomographic properties of forme fruste keratoconus corneas. *Invest Ophthalmol Vis Sci*. 2010;51:5546–5555.
14. RAs Jr, Alonso RS, Luz A, et al. Corneal-thickness spatial profile and corneal-volume distribution: tomographic indices to detect keratoconus. *J Cataract Refractive Surg*. 2006;32:1851–1859.
15. Oliveira CM, Ferreira A, Franco S. Wavefront analysis and Zernike polynomial decomposition for evaluation of corneal optical quality. *J Cataract Refract Surg*. 2012;38:343–356.
16. Schwiegerling J, Greivenkamp JE. Using corneal height maps and polynomial decomposition to determine corneal aberrations. *Optom Vis Sci*. 1997;74:906–916.
17. Martinez-Finkelshtein A, Delgado AM, Castro GM, et al. Comparative analysis of some modal reconstruction methods of the shape of the cornea from corneal elevation data. *Invest Ophthalmol Vis Sci*. 2009;50:5639–5645.
18. Applegate RA, Sarver EJ, Khemsara V. Are all aberrations equal? *Refract Surg*. 2002;18:556–562.
19. Kovács I, Miháltz K, Kránitz K, et al. Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. *J Cataract Refract Surg*. 2016;42:275–283.
20. Kuo BI, Chang WY, Liao TS, et al. Keratoconus screening based on deep learning approach of corneal topography. *Translational Vis Sci Tech*. 2020;9:53–57.
21. Consejo A, Melcer T, Rozema JJ. Introduction to machine learning for ophthalmologists. *Semin Ophthalmol*. 2018;34:19–41.
22. Gargeya R, Leng T. Automated identification of diabetic retinopathy using deep learning. *Ophthalmology*. 2017;124:962–969.
23. Abràmoff MD, Lou Y, Erginay A, et al. Improved automated detection of diabetic retinopathy on a publicly available dataset through integration of deep learning. *Invest Ophthalmol Vis Sci*. 2016;57:5200–5206.
24. Asaoka R, Murata H, Iwase A, et al. Detecting preperimetric glaucoma with standard automated perimetry using a deep learning classifier. *Ophthalmology*. 2016;123:1974–1980.
25. Lee CS, Baughman DM, Lee AY. Deep learning is effective for classifying normal versus Age-Related Macular Degeneration OCT images. *Ophthalmol Retina*. 2017;1:322–327.
26. Lee CS, Lee AY, Holland GN, et al. Big data and uveitis. *Ophthalmology*. 2016;123:2273–2275.
27. Bae GH, Kim JR, Kim CH, et al. Corneal topographic and tomographic analysis of fellow eyes in unilateral keratoconus patients using pentacam. *Am J Ophthalmol*. 2014;157:103–e1.
28. Vieira de Carvalho LA, Barbosa MS. Neural networks and statistical analysis for classification of corneal videokeratography maps based on Zernike coefficients: a quantitative comparison. *Arq Bras Oftalmol*. 2008;71:337–341.
29. Xie Y, Zhao L, Yang X, et al. Screening candidates for refractive surgery with corneal tomographic-based deep learning. *JAMA Ophthalmol*. 2020;138:519–526.
30. Zéboulon P, Debellemannière G, Gatinel D. Unsupervised learning for large-scale corneal topography clustering. *Sci Rep*. 2020;10:16973.
31. Lavric A, Valentin P. KeratoDetect: keratoconus detection algorithm using convolutional neural networks. *Comput Intell Neurosci*. 2019;2019:8162567.
32. Dos Santos VA, Schmetterer L, Stegmann H, et al. CorneaNet: fast segmentation of cornea OCT scans of healthy and keratoconic eyes using deep learning. *Biomed Opt Express*. 2019;10:622–641.
33. Arbelaez MC, Versaci F, Vestri G, et al. Use of a support vector machine for keratoconus and subclinical keratoconus detection by topographic and tomographic data. *Ophthalmology*. 2012;119:2231–2238.
34. Accardo PA, Pensierob S. Neural network-based system for early keratoconus detection from corneal topography. *J Biomed Inform*. 2003;35:151–159.
35. Souza MB, Medeiros FW, Souza DB, et al. Evaluation of machine learning classifiers in keratoconus detection from orbscan II examinations. *Clinics (Sao Paulo)*. 2010;65:1223–1228.
36. Gomes JA, Tan D, Rapuano CJ, et al. Global consensus on keratoconus and ectatic diseases. *Cornea*. 2015;34:359–369.
37. Schwiegerling J. Scaling Zernike expansion coefficients to different pupil sizes. *J Opt Soc Am A Opt Image Sci Vis*. 2002;19:1937–1945.
38. Campbell CE. Matrix method to find a new set of Zernike coefficients from an original set when the aperture radius is changed. *J Opt Soc Am A Opt Image Sci Vis*. 2003;20:209–217.
39. Raskutti G, Wainwright MJ, Yu B. Early stopping for non-parametric regression: an optimal data-dependent stopping rule. *J Machine Learn Res*. 2014;15:335–366.
40. Applegate RA, Ballentine C, Gross H, et al. Visual acuity as a function of Zernike mode and level of root mean square error. *Optom Vis Sci*. 2003;80:97–105.
41. Fan L, Xiong L, Zhang B, et al. Longitudinal and regional non-uniform remodeling of corneal epithelium after topography-guided FS-LASIK. *J Refract Surg*. 2019;35:88–95.
42. Bühren J, Kook D, Yoon G, et al. Detection of subclinical keratoconus by using corneal anterior and posterior surface aberrations and thickness spatial profiles. *Invest Ophthalmol Vis Sci*. 2010;51:3424–3432.
43. Carvalho LA. Accuracy of Zernike polynomials in characterizing optical aberrations and the corneal surface of the eye. *Invest Ophthalmol Vis Sci*. 2005;46:1915–1926.
44. Xu Z, Li W, Jiang J, et al. Characteristic of entire corneal topography and tomography for the detection of sub-clinical keratoconus with Zernike polynomials using Pentacam. *Sci Rep*. 2017;7:16486.
45. Shetty R, FRCS HM, Srivatsa P. A novel Zernike application to differentiate between 3-D corneal thickness of normal corneas and corneas with keratoconus. *Am J Ophthalmol*. 2015;160:453–462.
46. Hashemi H, Beiranvand A, Yekta A, et al. Pentacam top indices for diagnosing subclinical and definite keratoconus. *J Curr Ophthalmol*. 2016;28:21–26.
47. Issarti I, Consejo A, Jiménez-García M, et al. Computer aided diagnosis for suspect keratoconus detection. *Comput Biol Med*. 2019;109:33–42.
48. Oruçoglu F, Toker E. A Novel scoring system for distinguishing keratoconus from normal eyes. *Cont Lens Anterior Eye*. 2016;39:369–374.
49. Freegard TJ. The physical basis of transparency of the normal cornea. *Eye (Lond)*. 1997;11 (pt 4):465–471.
50. Reinstein DZ, Gobbe M, Archer TJ, et al. Epithelial, stromal, and total corneal thickness in keratoconus: three-dimensional display with artemis very-high frequency digital ultrasound. *J Refract Surg*. 2010;26:259–271.
51. Vega-Estrada A, Mimouni M, Espla E, et al. Corneal epithelial thickness intrasubject repeatability and its relation with visual limitation in keratoconus. *Ophthalmology*. 2019;200:255–262.